

MATH 535: LINEAR ALGEBRA

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1. AUGUST 24, 2026

1.1. Fields.

Definition 1 (Binary operation). A binary operation on a set S is a function

$$F: S \times S \longrightarrow S.$$

Thus, for every $x, y \in S$, the value $F(x, y)$ is again an element of S . We usually write $x \cdot y$ or $x + y$ in place of $F(x, y)$.

Example 1. Addition on $\mathbb{Z}$ is a binary operation: the map

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \quad (m, n) \longmapsto m + n$$

takes every pair of integers to an integer.

Definition 2. Let F be a binary operation on S . A subset $T \subseteq S$ is closed under F if

$$F(x, y) \in T$$

whenever $x, y \in T$.

Example 2. The positive integers $\mathbb{Z}_{>0}$ are closed under addition. The odd integers are not closed under addition, but they are closed under multiplication.

Definition 3 (Field). Let $\mathbb{F}$ be a set equipped with two binary operations, called addition and multiplication. The sum and product of $a, b \in \mathbb{F}$ are denoted by $a + b$ and ab , respectively. Suppose that for every $a, b, c \in \mathbb{F}$,

- (i) $a + b = b + a$;
- (ii) $(a + b) + c = a + (b + c)$;
- (iii) $ab = ba$;
- (iv) $a(bc) = (ab)c$;

$$(v) \ a(b + c) = ab + ac.$$

Suppose further that

- (vi) $\mathbb{F}$ contains an additive identity 0 and a multiplicative identity 1, so that

$$a + 0 = a \quad \text{and} \quad 1a = a$$

for every $a \in \mathbb{F}$;

- (vii) $1 \neq 0$;

- (viii) every $a \in \mathbb{F}$ has an additive inverse $-a \in \mathbb{F}$ such that $a + (-a) = 0$;

- (ix) every nonzero $a \in \mathbb{F}$ has a multiplicative inverse $a^{-1} \in \mathbb{F}$ such that $aa^{-1} = 1$.

Then $\mathbb{F}$ is called a field.

Example 3. The integers $\mathbb{Z}$ do not form a field: for example, 2 has no multiplicative inverse in $\mathbb{Z}$. The sets $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$, with their usual operations, are fields.

There are also finite fields. For example,

$$\mathbb{F}_2 = \{0, 1\} \quad \text{and} \quad \mathbb{F}_3 = \{0, 1, 2\},$$

where the elements are the possible remainders upon division by 2 and 3, respectively. Addition and multiplication in $\mathbb{F}_3$ are given by the following tables:

$$\begin{array}{c|ccc} + & 0 & 1 & 2 \\ \hline 0 & 0 & 1 & 2 \\ 1 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 \end{array} \quad \begin{array}{c|ccc} \cdot & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 2 \\ 2 & 0 & 2 & 1 \end{array}.$$

More generally, if p is prime, then

$$\mathbb{F}_p = \{0, 1, \dots, p-1\},$$

with addition and multiplication performed modulo p , is a field.

Lemma 1. If a is an element of a field $\mathbb{F}$, then $a \cdot 0 = 0$.

Proof. Since $0 + 0 = 0$, distributivity gives

$$a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding the additive inverse of $a \cdot 0$ to both sides yields $0 = a \cdot 0$. $\square$

2. AUGUST 26, 2026

2.1. Further properties of fields. The preceding lecture showed that $a \cdot 0 = 0$ for every element a of a field. We now obtain the zero-product property.

Proposition 1 (Zero-product property). *Let $a, b \in \mathbb{F}$. If $ab = 0$, then either $a = 0$ or $b = 0$.*

Proof. Suppose that $ab = 0$ and $a \neq 0$. Since $\mathbb{F}$ is a field, a^{-1} exists. Therefore

$$0 = a^{-1}0 = a^{-1}(ab) = (a^{-1}a)b = 1b = b.$$

Thus $a = 0$ or $b = 0$. $\square$

2.2. Characteristic of a field. For a positive integer n and $a \in \mathbb{F}$, write

$$na = \underbrace{a + a + \cdots + a}_{n \text{ terms}}.$$

Definition 4 (Characteristic). *Let $\mathbb{F}$ be a field. If $n1 = 0$ for some positive integer n , then the least such n is called the characteristic of $\mathbb{F}$, denoted $\text{char}(\mathbb{F})$. If $n1 \neq 0$ for every positive integer n , then $\mathbb{F}$ has characteristic zero.*

The fields $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ have characteristic zero. For every prime p , the field $\mathbb{F}_p$ has characteristic p . The rational function field $\mathbb{F}_p(t)$ is an infinite field of characteristic p .

Proposition 2. *If $\mathbb{F}$ has positive characteristic r , then r is prime.*

Proof. We have $r \neq 1$, since $1 \neq 0$. Suppose that r is composite, so

$$r = st$$

for integers s, t satisfying $1 < s, t < r$. Distributivity gives

$$(s1)(t1) = (st)1 = r1 = 0.$$

By the zero-product property, either $s1 = 0$ or $t1 = 0$. Both alternatives contradict the minimality of r . Hence r is prime. $\square$

Proposition 3. *If $\text{char}(\mathbb{F}) = p > 0$, then*

$$pa = 0$$

for every $a \in \mathbb{F}$.

Proof. Since $p1 = 0$, associativity and distributivity imply

$$pa = p(1a) = (p1)a = 0a = 0.$$

$\square$

Proposition 4. *Every finite field has positive characteristic.*

Proof. Let $\mathbb{F}$ be finite. The sequence

$$1, \quad 2 \cdot 1, \quad 3 \cdot 1, \quad \dots$$

cannot consist of distinct elements. Hence there are positive integers $m > n$ such that

$$m1 = n1.$$

Adding the additive inverse of $n1$ to both sides gives

$$(m - n)1 = 0.$$

Thus $\mathbb{F}$ has positive characteristic. $\square$

2.3. Algebraically closed fields.

Definition 5 (Algebraically closed field). *A field $\mathbb{F}$ is algebraically closed if every nonconstant polynomial $f \in \mathbb{F}[x]$ has a root in $\mathbb{F}$.*

The field $\mathbb{Q}$ is not algebraically closed, because $x^2 - 2$ has no root in $\mathbb{Q}$. The field $\mathbb{R}$ is not algebraically closed, because $x^2 + 1$ has no real root. By the fundamental theorem of algebra, $\mathbb{C}$ is algebraically closed.

Proposition 5. *No finite field is algebraically closed.*

Proof. Let

$$\mathbb{F} = \{a_1, \dots, a_q\}$$

be finite. Consider the nonconstant polynomial

$$f(x) = 1 + \prod_{i=1}^q (x - a_i).$$

For each $i = 1, \dots, q$,

$$f(a_i) = 1 \neq 0.$$

Thus f has no root in $\mathbb{F}$, so $\mathbb{F}$ is not algebraically closed. $\square$

2.4. Vectors and vector spaces. The spaces $\mathbb{R}^2$, $\mathbb{R}^3$, and, more generally,

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{R} \right\}$$

are the motivating examples of vector spaces.

Definition 6 (Vector space). *Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is a set V equipped with two operations,*

$$V \times V \longrightarrow V, \quad (u, v) \longmapsto u + v,$$

and

$$\mathbb{F} \times V \longrightarrow V, \quad (\alpha, v) \longmapsto \alpha v,$$

satisfying the following axioms for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{F}$:

$$(V1) \quad u + (v + w) = (u + v) + w;$$

$$(V2) \quad u + v = v + u;$$

$$(V3) \quad \text{there is an element } 0 \in V \text{ such that } v + 0 = 0 + v = v;$$

$$(V4) \quad \text{for every } v \in V, \text{ there is an element } -v \in V \text{ such that } v + (-v) = (-v) + v = 0;$$

$$(V5) \quad 1v = v;$$

$$(V6) \quad (\alpha\beta)v = \alpha(\beta v);$$

$$(V7) \quad \alpha(u + v) = \alpha u + \alpha v;$$

$$(V8) \quad (\alpha + \beta)v = \alpha v + \beta v.$$

The elements of V are called *vectors*, and the elements of $\mathbb{F}$ are called *scalars*.

3. AUGUST 28, 2026

3.1. Examples of vector spaces.

Example 4 (Coordinate spaces). *Let $\mathbb{F}$ be a field. The set*

$$\mathbb{F}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_1, \dots, x_n \in \mathbb{F} \right\}$$

is a vector space over $\mathbb{F}$ under componentwise addition and scalar multiplication:

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \quad \alpha \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \vdots \\ \alpha x_n \end{bmatrix}.$$

In particular, $\mathbb{R}^n$ is a vector space over $\mathbb{R}$ and $\mathbb{C}^n$ is a vector space over $\mathbb{C}$.

Example 5 (Matrix spaces). *The set $M_{m \times n}(\mathbb{F})$ of $m \times n$ matrices with entries in $\mathbb{F}$ is a vector space over $\mathbb{F}$. Addition and scalar multiplication are defined entrywise:*

$$[a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}], \quad \alpha[a_{ij}] = [\alpha a_{ij}].$$

Example 6 (Polynomial spaces). *Let*

$$\mathcal{P}_n(\mathbb{F}) = \{a_0 + a_1t + \dots + a_nt^n : a_0, \dots, a_n \in \mathbb{F}\}.$$

This is the vector space over $\mathbb{F}$ of polynomials of degree at most n . Its operations are

$$\begin{aligned} (a_0 + a_1t + \dots + a_nt^n) + (b_0 + b_1t + \dots + b_nt^n) \\ = (a_0 + b_0) + (a_1 + b_1)t + \dots + (a_n + b_n)t^n, \end{aligned}$$

and

$$\alpha(a_0 + a_1t + \dots + a_nt^n) = \alpha a_0 + \alpha a_1t + \dots + \alpha a_nt^n.$$

3.2. Linear combinations and spanning sets.

Definition 7 (Linear combination and span). *Let V be a vector space over $\mathbb{F}$ and let $v_1, \dots, v_p \in V$. A linear combination of these vectors is a vector of the form*

$$\alpha_1 v_1 + \dots + \alpha_p v_p, \quad \alpha_1, \dots, \alpha_p \in \mathbb{F}.$$

The set of all their linear combinations is called their span:

$$\text{span}\{v_1, \dots, v_p\} = \{\alpha_1 v_1 + \dots + \alpha_p v_p : \alpha_1, \dots, \alpha_p \in \mathbb{F}\}.$$

A linear combination is trivial if all of its coefficients are zero.

In $\mathbb{R}^2$, let

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then

$$\text{span}\{e_1, e_2\} = \mathbb{R}^2,$$

because

$$\begin{bmatrix} x \\ y \end{bmatrix} = x e_1 + y e_2.$$

If

$$v = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

then $\{e_1, e_2, v\}$ also spans $\mathbb{R}^2$, since

$$\begin{bmatrix} x \\ y \end{bmatrix} = x e_1 + y e_2 + 0 v.$$

The additional vector is unnecessary, which motivates the question of redundancy in a spanning set.

Definition 8 (Spanning set). *A collection $\{v_1, \dots, v_p\} \subseteq V$ is a spanning set for V if every $v \in V$ can be written as a linear combination*

$$v = \alpha_1 v_1 + \dots + \alpha_p v_p.$$

Equivalently,

$$\text{span}\{v_1, \dots, v_p\} = V.$$

3.3. Linear independence.

Definition 9 (Linear independence). *A collection $\{v_1, \dots, v_p\} \subseteq V$ is linearly independent if*

$$\alpha_1 v_1 + \dots + \alpha_p v_p = 0$$

implies

$$\alpha_1 = \dots = \alpha_p = 0.$$

In other words, only the trivial linear combination of $v_1, \dots, v_p$ equals zero. A collection that is not linearly independent is called linearly dependent.

Example 7. *Consider the vectors*

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

in $\mathbb{R}^3$. To test linear independence, solve

$$\alpha v_1 + \beta v_2 + \gamma v_3 = 0.$$

This gives the system

$$\begin{cases} \alpha + 2\beta + 3\gamma = 0, \\ \alpha + \gamma = 0, \\ \beta + \gamma = 0. \end{cases}$$

Taking $\gamma = 1$ gives $\alpha = \beta = -1$, so

$$-v_1 - v_2 + v_3 = 0.$$

Thus the vectors are linearly dependent; equivalently, $v_3 = v_1 + v_2$.

The following result precisely characterizes redundancy in a finite collection of vectors.

Proposition 6. *Let $v_1, \dots, v_p \in V$. The collection $\{v_1, \dots, v_p\}$ is linearly dependent if and only if one of its vectors is a linear combination of the others. Equivalently, there is an index k and scalars $\beta_j \in \mathbb{F}$ such*

that

$$v_k = \sum_{\substack{j=1 \\ j \neq k}}^p \beta_j v_j.$$

Proof. Suppose first that $\{v_1, \dots, v_p\}$ is linearly dependent. Then there are scalars $\alpha_1, \dots, \alpha_p$, not all zero, such that

$$\alpha_1 v_1 + \dots + \alpha_p v_p = 0.$$

Choose k such that $\alpha_k \neq 0$. Since $\mathbb{F}$ is a field, α_k^{-1} exists, and we can solve for v_k :

$$v_k = - \sum_{\substack{j=1 \\ j \neq k}}^p \alpha_k^{-1} \alpha_j v_j.$$

Thus v_k is a linear combination of the other vectors.

Conversely, suppose that for some k ,

$$v_k = \sum_{\substack{j=1 \\ j \neq k}}^p \beta_j v_j.$$

Moving all terms to one side gives

$$v_k - \sum_{\substack{j=1 \\ j \neq k}}^p \beta_j v_j = 0.$$

This is a nontrivial linear combination of $v_1, \dots, v_p$, because the coefficient of v_k is $1 \neq 0$. Hence the collection is linearly dependent. $\square$

4. AUGUST 31, 2026

4.1. Bases and coordinates.

Definition 10 (Basis). *A collection of vectors $\{v_1, \dots, v_n\}$ in a vector space V is a basis for V if it is linearly independent and spans V .*

The two conditions in the definition can be combined into a statement about representing vectors uniquely.

Proposition 7. *A collection $\{v_1, \dots, v_n\}$ is a basis for V if and only if every vector $v \in V$ has a unique representation of the form*

$$v = a_1v_1 + \dots + a_nv_n, \quad a_1, \dots, a_n \in \mathbb{F}.$$

When $\{v_1, \dots, v_n\}$ is a basis, the scalars $a_1, \dots, a_n$ are called the coordinates of v with respect to this basis.

Proof. Suppose first that $\{v_1, \dots, v_n\}$ is a basis. Since the collection spans V , every $v \in V$ has a representation

$$v = a_1v_1 + \dots + a_nv_n.$$

To prove uniqueness, suppose that

$$v = \tilde{a}_1v_1 + \dots + \tilde{a}_nv_n$$

is another such representation. Subtracting the two equations gives

$$(a_1 - \tilde{a}_1)v_1 + \dots + (a_n - \tilde{a}_n)v_n = 0.$$

Because $\{v_1, \dots, v_n\}$ is linearly independent,

$$a_i - \tilde{a}_i = 0$$

for each i . Thus $a_i = \tilde{a}_i$ for every i , and the representation is unique.

Conversely, suppose every $v \in V$ has a unique representation as a linear combination of $v_1, \dots, v_n$. The existence of such a representation for every v shows that the collection spans V . Now suppose

$$a_1v_1 + \dots + a_nv_n = 0.$$

The zero vector also has the representation

$$0 = 0v_1 + \dots + 0v_n.$$

Uniqueness therefore gives $a_1 = \dots = a_n = 0$. Hence the collection is linearly independent, so it is a basis for V . $\square$

Example 8 (The standard basis of $\mathbb{F}^n$). *For $i = 1, \dots, n$, let $e_i \in \mathbb{F}^n$ be the vector whose i th coordinate is 1 and whose other coordinates are*

0. *Thus*

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

Every vector in $\mathbb{F}^n$ has the unique representation

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 e_1 + x_2 e_2 + \dots + x_n e_n.$$

Consequently, $\{e_1, \dots, e_n\}$ is a basis for $\mathbb{F}^n$, called its standard basis.

Example 9 (The standard basis of $\mathcal{P}_n(\mathbb{F})$). Recall that $\mathcal{P}_n(\mathbb{F})$ is the vector space of polynomials over $\mathbb{F}$ of degree at most n . Each polynomial has a unique expression

$$p(t) = a_0 + a_1 t + \dots + a_n t^n.$$

Hence

$$\{1, t, t^2, \dots, t^n\}$$

is a basis for $\mathcal{P}_n(\mathbb{F})$, called its standard basis.

Proposition 8. Every finite nonempty spanning set for a vector space contains a basis.

Proof. Let $\{v_1, \dots, v_p\}$ span V . If this collection is linearly independent, then it is already a basis. Otherwise, by Proposition 6, some v_k is a linear combination of the other vectors. It follows that removing v_k does not change the span:

$$\text{span}\{v_1, \dots, v_p\} = \text{span}\{v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_p\}.$$

If the remaining collection is linearly independent, it is a basis. If not, repeat the same argument. The original spanning set is finite, so this process must terminate. The final collection still spans V and is

linearly independent, and is therefore a basis contained in the original spanning set. $\square$

4.2. Linear transformations. A *transformation*, also called a map or function, from a set X to a set Y is a rule $T: X \rightarrow Y$ that assigns to every $x \in X$ an element $T(x) \in Y$.

Definition 11 (Linear transformation). *Let V and W be vector spaces over the same field $\mathbb{F}$. A transformation $T: V \rightarrow W$ is linear if*

- (i) $T(u + v) = T(u) + T(v)$ for all $u, v \in V$;
- (ii) $T(\alpha u) = \alpha T(u)$ for all $u \in V$ and $\alpha \in \mathbb{F}$.

Equivalently, $T: V \rightarrow W$ is linear if

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$$

for all $u, v \in V$ and $\alpha, \beta \in \mathbb{F}$.

Example 10 (Reflection). *Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by*

$$T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix}.$$

Geometrically, T reflects each point across the first coordinate axis. To check additivity, observe that

$$\begin{aligned} T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right) &= T \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 + y_1 \\ -x_2 - y_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ -y_2 \end{bmatrix} \\ &= T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + T \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}. \end{aligned}$$

For $\alpha \in \mathbb{R}$,

$$\begin{aligned} T\left(\alpha \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) &= T \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \end{bmatrix} \\ &= \begin{bmatrix} \alpha x_1 \\ -\alpha x_2 \end{bmatrix} \\ &= \alpha \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix} = \alpha T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \end{aligned}$$

Thus T is linear.

Example 11 (The differentiation operator). Let $n \geq 1$, $V = \mathcal{P}_n(\mathbb{F})$, and $W = \mathcal{P}_{n-1}(\mathbb{F})$. Define

$$T: V \longrightarrow W, \quad T(p) = p'.$$

Since

$$(p + q)' = p' + q' \quad \text{and} \quad (\alpha p)' = \alpha p',$$

the differentiation operator T is linear.

5. SEPTEMBER 2, 2026

5.1. Linear transformations determined by a basis. The following result shows that prescribing the values of a linear transformation on a basis completely determines the transformation.

Lemma 2 (Linear map lemma). Suppose that $\{v_1, \dots, v_n\}$ is a basis for V , and let $w_1, \dots, w_n$ be arbitrary vectors in W . Then there exists a unique linear transformation $T: V \rightarrow W$ such that

$$T(v_k) = w_k$$

for each $k = 1, \dots, n$.

Proof. Every $v \in V$ has a unique representation

$$v = \sum_{i=1}^n c_i v_i.$$

Define

$$T(v) = \sum_{i=1}^n c_i w_i.$$

The uniqueness of the coordinates $c_1, \dots, c_n$ shows that this definition is well-defined. In particular, taking $v = v_k$ gives $T(v_k) = w_k$.

To prove that T is linear, write

$$u = \sum_{i=1}^n a_i v_i \quad \text{and} \quad v = \sum_{i=1}^n c_i v_i.$$

Then

$$\begin{aligned} T(u+v) &= T\left(\sum_{i=1}^n (a_i + c_i) v_i\right) \\ &= \sum_{i=1}^n (a_i + c_i) w_i \\ &= \sum_{i=1}^n a_i w_i + \sum_{i=1}^n c_i w_i \\ &= T(u) + T(v). \end{aligned}$$

Similarly, for every $\alpha \in \mathbb{F}$,

$$\begin{aligned} T(\alpha v) &= T\left(\sum_{i=1}^n \alpha c_i v_i\right) \\ &= \sum_{i=1}^n \alpha c_i w_i = \alpha \sum_{i=1}^n c_i w_i = \alpha T(v). \end{aligned}$$

Thus T is linear.

It remains to prove uniqueness. Suppose that $T': V \rightarrow W$ is another linear transformation satisfying $T'(v_k) = w_k$ for every k . If $v = \sum_{i=1}^n c_i v_i$, then

$$T'(v) = \sum_{i=1}^n c_i T'(v_i) = \sum_{i=1}^n c_i w_i = T(v).$$

Hence $T' = T$. □

5.2. Matrix representations of linear transformations. An *ordered basis* is a basis whose vectors are listed in a fixed order. Let

$$\mathcal{B} = (v_1, \dots, v_n) \quad \text{and} \quad \mathcal{C} = (w_1, \dots, w_m)$$

be ordered bases for V and W , respectively. For $v = x_1v_1 + \dots + x_nv_n$, write

$$[v]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

for the coordinate vector of v with respect to $\mathcal{B}$.

Theorem 1 (Matrix representation theorem). *Let V and W be vector spaces over $\mathbb{F}$ with ordered bases $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_m)$. For every linear transformation $T: V \rightarrow W$, there is a unique matrix*

$$A = [A_{ij}] = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \in M_{m \times n}(\mathbb{F})$$

such that

$$T(v_j) = \sum_{i=1}^m A_{ij}w_i$$

for each $j = 1, \dots, n$. Conversely, every matrix $A \in M_{m \times n}(\mathbb{F})$ determines a unique linear transformation $T: V \rightarrow W$ through this formula.

Proof. Let $T: V \rightarrow W$ be linear. Because $\mathcal{C}$ is a basis for W , each vector $T(v_j)$ has a unique representation

$$T(v_j) = A_{1j}w_1 + \cdots + A_{mj}w_m = \sum_{i=1}^m A_{ij}w_i.$$

Use these coordinate vectors as the columns of A :

$$A = \begin{bmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \cdots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{bmatrix}.$$

This constructs the required matrix. If $B = [B_{ij}]$ were another matrix with the same property, then for every j ,

$$\sum_{i=1}^m B_{ij}w_i = T(v_j) = \sum_{i=1}^m A_{ij}w_i.$$

The uniqueness of coordinates with respect to $\mathcal{C}$ gives $B_{ij} = A_{ij}$ for every i and j . Hence $B = A$.

Conversely, let $A = [A_{ij}]$ be an arbitrary $m \times n$ matrix and define

$$\tilde{w}_j = \sum_{i=1}^m A_{ij}w_i \quad (j = 1, \dots, n).$$

By the linear map lemma, there is a unique linear transformation $T: V \rightarrow W$ such that $T(v_j) = \tilde{w}_j$ for every j . Thus A determines a unique linear transformation with the required property. $\square$

If

$$v = x_1v_1 + \dots + x_nv_n,$$

then linearity gives

$$\begin{aligned} T(v) &= \sum_{j=1}^n x_j T(v_j) \\ &= \sum_{j=1}^n x_j \sum_{i=1}^m A_{ij}w_i \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n A_{ij}x_j \right) w_i. \end{aligned}$$

Consequently,

$$[T(v)]_{\mathcal{C}} = A[v]_{\mathcal{B}}.$$

Thus multiplication by the matrix A computes the coordinates of the image of a vector from the coordinates of the original vector.

6. SEPTEMBER 4, 2026

6.1. Matrix–vector multiplication. Let V and W be vector spaces over $\mathbb{F}$ with ordered bases

$$\mathcal{B} = (v_1, \dots, v_n) \quad \text{and} \quad \mathcal{C} = (w_1, \dots, w_m),$$

respectively, and let $T: V \rightarrow W$ be linear. Recall that the matrix representing T with respect to these bases is

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix},$$

where

$$T(v_j) = A_{1j}w_1 + \cdots + A_{mj}w_m.$$

Thus the j th column of A is the coordinate vector $[T(v_j)]_C$.

Suppose that

$$v = x_1v_1 + \cdots + x_nv_n, \quad [v]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

By linearity,

$$\begin{aligned} [T(v)]_C &= x_1[T(v_1)]_C + \cdots + x_n[T(v_n)]_C \\ &= x_1 \begin{bmatrix} A_{11} \\ A_{21} \\ \vdots \\ A_{m1} \end{bmatrix} + \cdots + x_n \begin{bmatrix} A_{1n} \\ A_{2n} \\ \vdots \\ A_{mn} \end{bmatrix} \\ &= \begin{bmatrix} A_{11}x_1 + \cdots + A_{1n}x_n \\ \vdots \\ A_{m1}x_1 + \cdots + A_{mn}x_n \end{bmatrix}. \end{aligned}$$

This motivates the following definition.

Definition 12 (Matrix–vector product). *Let $A = [A_{ij}] \in M_{m \times n}(\mathbb{F})$, and write its columns as $A_1, \dots, A_n$. For*

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{F}^n,$$

define

$$Ax = x_1A_1 + \cdots + x_nA_n.$$

Equivalently, the k th coordinate of Ax is

$$(Ax)_k = \sum_{j=1}^n A_{kj}x_j, \quad k = 1, \dots, m.$$

With this notation, the coordinate formula for a linear transformation is

$$[T(v)]_C = A[v]_B.$$

Example 12. *Matrix–vector multiplication can be computed either as a linear combination of the columns or one coordinate at a time. For instance,*

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 14 \\ 10 \end{bmatrix},$$

while

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 \\ 4 \cdot 1 + 5 \cdot 2 + 6 \cdot 3 \end{bmatrix} = \begin{bmatrix} 14 \\ 32 \end{bmatrix}.$$

6.2. Finding matrix representations.

Example 13. *Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by*

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 2y \\ 2x - 5y \\ 7y \end{bmatrix}.$$

With respect to the standard bases, the columns of the representing matrix are

$$T(e_1) = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \quad \text{and} \quad T(e_2) = \begin{bmatrix} 2 \\ -5 \\ 7 \end{bmatrix}.$$

Hence

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -5 \\ 0 & 7 \end{bmatrix}.$$

Example 14 (Differentiation). *Let $n \geq 1$, and let*

$$T: \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R}), \quad T(p) = p',$$

and use the ordered basis

$$\mathcal{B} = (1, t, t^2, \dots, t^n)$$

for both the domain and codomain. Since

$$T(1) = 0, \quad T(t) = 1, \quad T(t^j) = jt^{j-1},$$

the matrix representing T is

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & n \\ 0 & \cdots & \cdots & 0 & 0 \end{bmatrix}.$$

6.3. The vector space of linear maps. Let V and W be vector spaces over $\mathbb{F}$. If $T: V \rightarrow W$ is linear and $\alpha \in \mathbb{F}$, define $\alpha T: V \rightarrow W$ pointwise by

$$(\alpha T)(v) = \alpha T(v).$$

This map is again linear. Indeed, for $u, v \in V$,

$$\begin{aligned} (\alpha T)(u + v) &= \alpha T(u + v) \\ &= \alpha(T(u) + T(v)) \\ &= (\alpha T)(u) + (\alpha T)(v), \end{aligned}$$

and, for $\beta \in \mathbb{F}$,

$$(\alpha T)(\beta u) = \alpha T(\beta u) = \alpha\beta T(u) = \beta(\alpha T)(u).$$

Similarly, if $T_1, T_2: V \rightarrow W$ are linear, define their sum by

$$(T_1 + T_2)(v) = T_1(v) + T_2(v).$$

The map $T_1 + T_2$ is linear because

$$(T_1 + T_2)(u + v) = T_1(u + v) + T_2(u + v)$$

$$= (T_1 + T_2)(u) + (T_1 + T_2)(v)$$

and

$$(T_1 + T_2)(\alpha u) = \alpha(T_1 + T_2)(u).$$

Definition 13 (Space of linear maps). *The collection of all linear transformations from V to W is denoted by*

$$\mathcal{L}(V, W) = \{T: V \rightarrow W : T \text{ is linear}\}.$$

Proposition 9. *The set $\mathcal{L}(V, W)$, equipped with pointwise addition and scalar multiplication, is a vector space over $\mathbb{F}$.*

Proof. The preceding calculations show that the two operations are closed in $\mathcal{L}(V, W)$. All vector-space identities follow pointwise from the corresponding identities in W . The additive identity is the zero transformation $0: V \rightarrow W$, defined by $0(v) = 0$, and the additive inverse of T is the map $-T$, defined by $(-T)(v) = -T(v)$. $\square$

6.4. Composition of linear maps. Let V , W , and Z be vector spaces over $\mathbb{F}$. Suppose that

$$T_2: V \longrightarrow W \quad \text{and} \quad T_1: W \longrightarrow Z$$

are linear. Their *composition* is the map

$$T_1 \circ T_2: V \longrightarrow Z, \quad (T_1 \circ T_2)(v) = T_1(T_2(v)).$$

Proposition 10. *The composition $T_1 \circ T_2$ is linear.*

Proof. For $u, v \in V$ and $\alpha \in \mathbb{F}$,

$$\begin{aligned} (T_1 \circ T_2)(u + v) &= T_1(T_2(u) + T_2(v)) \\ &= T_1(T_2(u)) + T_1(T_2(v)), \end{aligned}$$

and

$$(T_1 \circ T_2)(\alpha u) = T_1(\alpha T_2(u)) = \alpha T_1(T_2(u)).$$

$\square$

7. SEPTEMBER 9, 2026

7.1. Matrix operations and linear maps. Let V and W be vector spaces over $\mathbb{F}$ with ordered bases

$$\mathcal{B} = (v_1, \dots, v_n) \quad \text{and} \quad \mathcal{C} = (w_1, \dots, w_m).$$

Suppose that $T, S \in \mathcal{L}(V, W)$ are represented by the matrices $A = [A_{ij}]$ and $B = [B_{ij}]$, respectively. Thus

$$T(v_j) = \sum_{i=1}^m A_{ij} w_i \quad \text{and} \quad S(v_j) = \sum_{i=1}^m B_{ij} w_i.$$

For each basis vector v_j ,

$$\begin{aligned} (T + S)(v_j) &= T(v_j) + S(v_j) \\ &= \sum_{i=1}^m A_{ij} w_i + \sum_{i=1}^m B_{ij} w_i \\ &= \sum_{i=1}^m (A_{ij} + B_{ij}) w_i. \end{aligned}$$

Consequently, the matrix representing $T + S$ is obtained by adding the matrices representing T and S entrywise.

Definition 14 (Matrix addition). *If $A, B \in M_{m \times n}(\mathbb{F})$, their sum is the matrix $A + B \in M_{m \times n}(\mathbb{F})$ defined by*

$$(A + B)_{ij} = A_{ij} + B_{ij}.$$

Similarly, if $\alpha \in \mathbb{F}$, then

$$(\alpha T)(v_j) = \alpha T(v_j) = \sum_{i=1}^m \alpha A_{ij} w_i.$$

Thus scalar multiplication of linear maps corresponds to entrywise scalar multiplication of matrices.

Definition 15 (Scalar multiplication of matrices). *If $A \in M_{m \times n}(\mathbb{F})$ and $\alpha \in \mathbb{F}$, then $\alpha A \in M_{m \times n}(\mathbb{F})$ is defined by*

$$(\alpha A)_{ij} = \alpha A_{ij}.$$

7.2. Composition and matrix multiplication. Let Z be a third vector space with ordered basis

$$\mathcal{D} = (z_1, \dots, z_p),$$

and consider linear maps

$$T_2: V \longrightarrow W \quad \text{and} \quad T_1: W \longrightarrow Z.$$

Suppose T_2 is represented by $B = [B_{kj}] \in M_{m \times n}(\mathbb{F})$ and T_1 is represented by $A = [A_{ik}] \in M_{p \times m}(\mathbb{F})$. For $j = 1, \dots, n$,

$$\begin{aligned} (T_1 \circ T_2)(v_j) &= T_1 \left(\sum_{k=1}^m B_{kj} w_k \right) \\ &= \sum_{k=1}^m B_{kj} T_1(w_k) \\ &= \sum_{k=1}^m B_{kj} \sum_{i=1}^p A_{ik} z_i \\ &= \sum_{i=1}^p \left(\sum_{k=1}^m A_{ik} B_{kj} \right) z_i. \end{aligned}$$

The coefficient of z_i in this expression is the (i, j) entry of the matrix representing $T_1 \circ T_2$.

Definition 16 (Matrix multiplication). *If $A \in M_{p \times m}(\mathbb{F})$ and $B \in M_{m \times n}(\mathbb{F})$, their product is the matrix $AB \in M_{p \times n}(\mathbb{F})$ whose entries are*

$$(AB)_{ij} = \sum_{k=1}^m A_{ik} B_{kj}.$$

Thus composition of linear maps corresponds to multiplication of their representing matrices, in the same order:

$$[T_1 \circ T_2] = [T_1][T_2].$$

Whenever all the products and sums involved are defined, matrix multiplication satisfies

$$(AB)C = A(BC),$$

$$A(B + C) = AB + AC,$$

$$(A + B)C = AC + BC,$$

$$A(\alpha B) = \alpha(AB) = (\alpha A)B.$$

In general, however, $AB \neq BA$; the two products may even have different sizes, or one may be undefined. In particular, $M_{n \times n}(\mathbb{F})$, with matrix addition, scalar multiplication, and matrix multiplication, is an associative algebra over $\mathbb{F}$.

7.3. Vector subspaces.

Definition 17 (Subspace). *Let V be a vector space over $\mathbb{F}$. A subset $U \subseteq V$ is a subspace of V if U , with the addition and scalar multiplication inherited from V , is itself a vector space over $\mathbb{F}$.*

Checking all the vector-space axioms is unnecessary. The following criterion reduces the verification to three conditions.

Proposition 11 (Subspace criterion). *A subset $U \subseteq V$ is a subspace of V if and only if*

- (i) $0 \in U$;
- (ii) $u + v \in U$ whenever $u, v \in U$;
- (iii) $\alpha u \in U$ whenever $u \in U$ and $\alpha \in \mathbb{F}$.

Equivalently, if U is known to be nonempty, it is enough to check the two closure conditions (ii) and (iii).

Example 15. *The affine line*

$$H = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 : x = 2 \right\}$$

is not a subspace of $\mathbb{R}^2$, because it does not contain the zero vector.

The subspaces of $\mathbb{R}^2$ are precisely the zero subspace $\{0\}$, the lines through the origin, and $\mathbb{R}^2$ itself.

8. SEPTEMBER 11, 2026

8.1. The subspace criterion.

Proof of Proposition 11. If U is a subspace of V , then U is a vector space under the operations inherited from V . It therefore contains the additive identity and is closed under addition and scalar multiplication.

Conversely, suppose conditions (i)–(iii) hold. The additive identity belongs to U by (i). If $u \in U$, then condition (iii), applied with the scalar -1 , gives

$$-u = (-1)u \in U.$$

Hence U contains the additive inverse of each of its elements. The remaining vector-space axioms are identities already valid in V , so they remain valid when the vectors are restricted to U . Thus U is a subspace of V .

Finally, if U is nonempty and satisfies the two closure conditions, choose $u \in U$. Closure under scalar multiplication gives $0 = 0u \in U$, so condition (i) follows automatically. $\square$

8.2. Kernels and images.

Definition 18 (Kernel and image). *Let $T \in \mathcal{L}(V, W)$. The kernel and image of T are*

$$\ker T = \{v \in V : T(v) = 0\}$$

and

$$\operatorname{im} T = \{T(v) : v \in V\},$$

respectively. The image is also called the range of T .

Proposition 12. *If $T \in \mathcal{L}(V, W)$, then $\ker T$ is a subspace of V and $\operatorname{im} T$ is a subspace of W .*

Proof. Since $T(0) = 0$, both sets contain the appropriate zero vector. If $u, v \in \ker T$ and $\alpha \in \mathbb{F}$, then

$$T(u + v) = T(u) + T(v) = 0 \quad \text{and} \quad T(\alpha u) = \alpha T(u) = 0.$$

Thus $\ker T$ is a subspace. If $y_1 = T(v_1)$ and $y_2 = T(v_2)$ are in $\operatorname{im} T$, then

$$y_1 + y_2 = T(v_1 + v_2) \quad \text{and} \quad \alpha y_1 = T(\alpha v_1),$$

so the image is also a subspace. $\square$

Example 16. *The continuous real-valued functions on $[0, 1]$ form a subspace*

$$C([0, 1]) \subseteq \{f : [0, 1] \rightarrow \mathbb{R}\}$$

of the vector space of all real-valued functions on $[0, 1]$.

Example 17. *Fix $b \in \mathbb{F}$ and let*

$$U_b = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{F}^4 : x_3 = 5x_4 + b \right\}.$$

Then U_b is a subspace of $\mathbb{F}^4$ if and only if $b = 0$. Indeed, the zero vector belongs to U_b only when $b = 0$, and when $b = 0$ the defining equation $x_3 = 5x_4$ is preserved by addition and scalar multiplication.

8.3. Sums of subspaces.

Definition 19 (Sum of subspaces). *If $V_1, \dots, V_m$ are subspaces of V , their sum is*

$$V_1 + \dots + V_m = \{v_1 + \dots + v_m : v_i \in V_i \text{ for } i = 1, \dots, m\}.$$

Example 18. *Let*

$$U = \left\{ \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} : x \in \mathbb{F} \right\}, \quad W = \left\{ \begin{bmatrix} 0 \\ y \\ 0 \end{bmatrix} : y \in \mathbb{F} \right\}.$$

Then

$$U + W = \left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} : x, y \in \mathbb{F} \right\}.$$

Proposition 13. *If $V_1, \dots, V_m$ are subspaces of V , then $V_1 + \dots + V_m$ is the smallest subspace of V containing each V_i .*

Proof. The sum contains $0 = 0 + \dots + 0$. Adding two elements of the sum, or multiplying one by a scalar, can be done componentwise inside each V_i , so the sum is a subspace. For each i , any $v_i \in V_i$ can be written

as

$$v_i = 0 + \cdots + 0 + v_i + 0 + \cdots + 0,$$

and hence $V_i \subseteq V_1 + \cdots + V_m$.

Conversely, let W be any subspace of V containing all the V_i . If $v = v_1 + \cdots + v_m$ with $v_i \in V_i$, then every v_i belongs to W . Since W is closed under addition, $v \in W$. Therefore $V_1 + \cdots + V_m \subseteq W$. $\square$

8.4. Direct sums.

Definition 20 (Direct sum). *Let $V_1, \dots, V_m$ be subspaces of V . Their sum is a direct sum if every element of $V_1 + \cdots + V_m$ has a unique representation*

$$v = v_1 + \cdots + v_m, \quad v_i \in V_i.$$

In this case we write

$$V_1 \oplus \cdots \oplus V_m.$$

Proposition 14 (Zero-sum criterion). *The sum $V_1 + \cdots + V_m$ is direct if and only if the equation*

$$0 = v_1 + \cdots + v_m, \quad v_i \in V_i,$$

implies $v_1 = \cdots = v_m = 0$.

Proof. If the sum is direct, then $0 = 0 + \cdots + 0$ is one representation of the zero vector, so uniqueness forces every other representation to have all components equal to zero.

Conversely, suppose the zero vector has only the trivial representation. If

$$v_1 + \cdots + v_m = u_1 + \cdots + u_m,$$

where $v_i, u_i \in V_i$, then

$$0 = (v_1 - u_1) + \cdots + (v_m - u_m).$$

The hypothesis gives $v_i - u_i = 0$ for each i , so $v_i = u_i$. Thus every representation is unique. $\square$

Proposition 15. *If U and W are subspaces of V , then*

$$U + W = U \oplus W \iff U \cap W = \{0\}.$$

Proof. Suppose first that $U + W$ is a direct sum, and let $v \in U \cap W$. Then

$$0 = v + (-v) = 0 + 0$$

gives two representations with one term in U and one term in W . Uniqueness implies $v = 0$.

Conversely, assume $U \cap W = \{0\}$. If

$$u + w = u' + w', \quad u, u' \in U, \quad w, w' \in W,$$

then

$$u - u' = w' - w \in U \cap W.$$

Hence $u = u'$ and $w = w'$, so the representation is unique. $\square$

For the coordinate-axis subspaces in the preceding example, $U \cap W = \{0\}$, and hence their sum is $U \oplus W$.

8.5. Product vector spaces.

Definition 21 (Product vector space). *Let $V_1, \dots, V_m$ be vector spaces over $\mathbb{F}$. Their Cartesian product is*

$$V_1 \times \cdots \times V_m = \{(v_1, \dots, v_m) : v_i \in V_i\}.$$

Addition and scalar multiplication are defined coordinatewise:

$$\begin{aligned} (v_1, \dots, v_m) + (u_1, \dots, u_m) &= (v_1 + u_1, \dots, v_m + u_m), \\ \alpha(v_1, \dots, v_m) &= (\alpha v_1, \dots, \alpha v_m). \end{aligned}$$

With these operations, $V_1 \times \cdots \times V_m$ is a vector space over $\mathbb{F}$; all vector-space axioms hold coordinatewise.

Example 19. *In $\mathcal{P}_5(\mathbb{R}) \times \mathbb{R}^3$,*

$$\begin{aligned} \left(5 - 6t + 4t^2, \begin{bmatrix} 3 \\ 8 \\ 7 \end{bmatrix} \right) + \left(t + 9t^5, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \right) \\ = \left(5 - 5t + 4t^2 + 9t^5, \begin{bmatrix} 5 \\ 10 \\ 9 \end{bmatrix} \right). \end{aligned}$$

9. SEPTEMBER 14, 2026

9.1. Direct sums and injectivity. Let $V_1, \dots, V_m$ be subspaces of a vector space V . Define the summation map

$$\Sigma: V_1 \times \cdots \times V_m \longrightarrow V_1 + \cdots + V_m$$

by

$$\Sigma(v_1, \dots, v_m) = v_1 + \cdots + v_m.$$

This map is linear and surjective.

Proposition 16. *The sum $V_1 + \cdots + V_m$ is direct if and only if the summation map Σ is injective.*

Proof. The map Σ is injective precisely when

$$\Sigma(v_1, \dots, v_m) = \Sigma(u_1, \dots, u_m)$$

implies

$$(v_1, \dots, v_m) = (u_1, \dots, u_m).$$

Equivalently, every vector in $V_1 + \cdots + V_m$ has only one representation as a sum of vectors from the respective subspaces. This is exactly the definition of a direct sum. $\square$

9.2. Quotient vector spaces. Let U be a subset of V . For $v \in V$, define

$$v + U = \{v + u : u \in U\}.$$

The set $v + U$ is called a *translate* of U . When U is a subspace, a translate is also called a *coset* of U .

Example 20. *Let*

$$U = \left\{ \begin{bmatrix} x \\ 2x \end{bmatrix} : x \in \mathbb{R} \right\} \subseteq \mathbb{R}^2.$$

Then

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} + U = \left\{ \begin{bmatrix} 2+x \\ 1+2x \end{bmatrix} : x \in \mathbb{R} \right\}.$$

The original subspace is the line $y = 2x$, while this translate is the parallel line $y = 2x - 3$.

Definition 22 (Quotient space). *If U is a subspace of V , the quotient space of V by U is the set of all cosets of U :*

$$V/U = \{v + U : v \in V\}.$$

For the subspace $U \subseteq \mathbb{R}^2$ in the preceding example, $\mathbb{R}^2/U$ is the collection of all lines in $\mathbb{R}^2$ with slope 2.

Proposition 17. *Let U be a subspace of V and let $v, w \in V$. The following are equivalent:*

- (i) $v - w \in U$;
- (ii) $v + U = w + U$;
- (iii) $(v + U) \cap (w + U) \neq \emptyset$.

Proof. Suppose first that $v - w \in U$. If $v + u \in v + U$, then

$$v + u = w + (v - w + u) \in w + U,$$

because $v - w + u \in U$. Thus $v + U \subseteq w + U$. Since $w - v = -(v - w) \in U$, the reverse inclusion follows in the same way, so $v + U = w + U$.

If $v + U = w + U$, then the two cosets plainly have nonempty intersection.

Finally, suppose $(v + U) \cap (w + U) \neq \emptyset$. There are $u_1, u_2 \in U$ such that

$$v + u_1 = w + u_2.$$

Therefore

$$v - w = u_2 - u_1 \in U.$$

□

Thus two cosets of a subspace are either equal or disjoint. We define addition and scalar multiplication on V/U by

$$(v + U) + (w + U) = (v + w) + U,$$

$$\alpha(v + U) = \alpha v + U.$$

Proposition 17 shows that these operations do not depend on the choice of representatives. With these operations, V/U is a vector space over $\mathbb{F}$, whose zero vector is the coset $U = 0 + U$.

9.3. Finite-dimensional vector spaces.

Definition 23 (Finite-dimensional vector space). *A vector space V is finite-dimensional if some finite collection of vectors spans V . It is infinite-dimensional if it is not finite-dimensional.*

Example 21. *The coordinate space $\mathbb{F}^n$ is finite-dimensional because*

$$\mathbb{F}^n = \text{span}\{e_1, \dots, e_n\}.$$

Likewise, the space of polynomials of degree at most m is finite-dimensional because

$$\mathcal{P}_m(\mathbb{F}) = \text{span}\{1, t, t^2, \dots, t^m\}.$$

Example 22. *The vector space $\mathcal{P}(\mathbb{F})$ of all polynomials with coefficients in $\mathbb{F}$ is infinite-dimensional. Indeed, suppose a finite collection of polynomials spanned $\mathcal{P}(\mathbb{F})$, and let m be the largest degree among the nonzero polynomials in the collection. Every polynomial in their span would have degree at most m , so t^{m+1} could not belong to the span, a contradiction.*

10. SEPTEMBER 16, 2026

10.1. Comparing independent and spanning collections.

Proposition 18. *Suppose that $u_1, \dots, u_m$ is a linearly independent collection in V and that $w_1, \dots, w_n$ spans V . Then*

$$m \leq n.$$

Thus, in a finite-dimensional vector space, every linearly independent collection has cardinality at most that of every finite spanning collection.

Proof. We successively insert the vectors $u_1, u_2, \dots$ into the spanning list $w_1, \dots, w_n$, removing one of the w 's after each insertion.

First adjoin u_1 to obtain

$$u_1, w_1, \dots, w_n.$$

This collection is linearly dependent because $w_1, \dots, w_n$ spans V . Since $u_1 \neq 0$, one of the w 's is a linear combination of the other vectors. Remove that vector. The resulting collection still spans V and has n vectors.

Inductively, suppose that after $k-1$ steps we have a spanning collection of n vectors consisting of $u_1, \dots, u_{k-1}$ and $n-k+1$ of the original w 's. Adjoin u_k . The resulting collection is dependent. In a nontrivial dependence relation, some remaining w must have a nonzero coefficient; otherwise the relation would contradict the linear independence of $u_1, \dots, u_k$. We can therefore remove that w without changing the span.

If $m > n$, then after n steps the collection $u_1, \dots, u_n$ would span V . In particular, u_{n+1} would lie in their span, contradicting the linear independence of $u_1, \dots, u_m$. Hence $m \leq n$. $\square$

Example 23. *No collection of four vectors in $\mathbb{R}^3$ is linearly independent, because the three standard basis vectors span $\mathbb{R}^3$. Similarly, no collection of three vectors can span $\mathbb{R}^4$, because the four standard basis vectors are linearly independent in $\mathbb{R}^4$.*

10.2. Subspaces of finite-dimensional spaces.

Proposition 19. *Every subspace of a finite-dimensional vector space is finite-dimensional.*

Proof. Let U be a subspace of a finite-dimensional vector space V . If $U = \{0\}$, then the empty collection spans U , so U is finite-dimensional.

Otherwise choose a nonzero vector $u_1 \in U$. If $U = \text{span}\{u_1\}$, we are done. If not, choose

$$u_2 \in U \setminus \text{span}\{u_1\}.$$

Continue in this fashion: once $u_1, \dots, u_{k-1}$ have been chosen, stop if they span U ; otherwise choose

$$u_k \in U \setminus \text{span}\{u_1, \dots, u_{k-1}\}.$$

At every stage the chosen collection is linearly independent. Since V has a finite spanning collection, Proposition 18 bounds the number

of vectors in any linearly independent collection in V . The process must therefore stop after finitely many steps, at which point the chosen vectors span U . $\square$

Corollary 1. *Every finite-dimensional vector space has a basis.*

Proof. By definition, a finite-dimensional vector space has a finite spanning collection. Removing redundant vectors from that collection produces a basis. $\square$

10.3. Extending a linearly independent collection.

Proposition 20 (Basis extension theorem). *Every finite linearly independent collection in a finite-dimensional vector space can be extended to a basis of the space.*

Proof. Let $u_1, \dots, u_m$ be linearly independent in V , and choose a finite spanning collection $w_1, \dots, w_n$ for V . The combined collection

$$u_1, \dots, u_m, w_1, \dots, w_n$$

spans V . Examine the w 's in order. Whenever a vector w_j lies in the span of the vectors preceding it, remove it. Removing such a vector does not change the span, and none of the u_i is removed. After all the w 's have been considered, the remaining collection spans V and is linearly independent. It is therefore a basis containing $u_1, \dots, u_m$. $\square$

Example 24. *Consider the linearly independent vectors*

$$v_1 = \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -9 \\ 6 \\ 8 \end{bmatrix}$$

in $\mathbb{R}^3$. Adjoin the standard basis vectors e_1, e_2, e_3 . Since

$$e_1 = -\frac{1}{5}(v_2 - 2v_1),$$

the vector e_1 is redundant. On the other hand, e_2 is not in $\text{span}\{v_1, v_2\}$, so we retain it. Finally,

$$e_3 = -\frac{1}{10} \left(v_2 - \frac{9}{2}v_1 + \frac{15}{2}e_2 \right),$$

so e_3 is redundant. Therefore

$$\left\{ \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -9 \\ 6 \\ 8 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

is a basis of $\mathbb{R}^3$ extending the original collection.

11. SEPTEMBER 18, 2026

11.1. Complements of subspaces.

Proposition 21. *Suppose that V is finite-dimensional and that U is a subspace of V . Then there is a subspace W of V such that*

$$V = U \oplus W.$$

Proof. Because V is finite-dimensional, so is U . Let $u_1, \dots, u_m$ be a basis of U . This collection is linearly independent in V , so the basis extension theorem gives a basis

$$u_1, \dots, u_m, w_1, \dots, w_n$$

of V . Let

$$W = \text{span}\{w_1, \dots, w_n\}.$$

Certainly $U + W \subseteq V$. Conversely, if $v \in V$, then there are scalars $a_1, \dots, a_m, b_1, \dots, b_n$ such that

$$v = \underbrace{a_1u_1 + \dots + a_mu_m}_{\in U} + \underbrace{b_1w_1 + \dots + b_nw_n}_{\in W}.$$

Hence $v \in U + W$, and therefore $V = U + W$.

It remains to show that $U \cap W = \{0\}$. Let $v \in U \cap W$. Then

$$v = a_1u_1 + \dots + a_mu_m = b_1w_1 + \dots + b_nw_n$$

for some scalars a_i and b_j . Thus

$$a_1u_1 + \dots + a_mu_m - b_1w_1 - \dots - b_nw_n = 0.$$

Because the combined collection is a basis of V , all the coefficients are zero. In particular, $v = 0$, so $U \cap W = \{0\}$. Thus $V = U \oplus W$. $\square$

11.2. Dimension.

Proposition 22. *Any two bases of a finite-dimensional vector space have the same cardinality.*

Proof. Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be bases of a finite-dimensional vector space V . Because $\mathcal{B}_1$ is linearly independent and $\mathcal{B}_2$ spans V , Proposition 18 gives

$$|\mathcal{B}_1| \leq |\mathcal{B}_2|.$$

Interchanging the roles of the two bases gives

$$|\mathcal{B}_2| \leq |\mathcal{B}_1|.$$

Therefore $|\mathcal{B}_1| = |\mathcal{B}_2|$. □

Definition 24 (Dimension). *The dimension of a finite-dimensional vector space V is the cardinality of any basis of V . It is denoted by $\dim V$.*

Example 25. *The standard basis $\{e_1, \dots, e_n\}$ of $\mathbb{F}^n$ has n elements, so*

$$\dim \mathbb{F}^n = n.$$

Example 26. *The standard basis $\{1, x, \dots, x^m\}$ of $\mathcal{P}_m(\mathbb{F})$ has $m + 1$ elements, so*

$$\dim \mathcal{P}_m(\mathbb{F}) = m + 1.$$

Example 27. *Let*

$$U = \left\{ \begin{bmatrix} x \\ 0 \\ y \end{bmatrix} : x, y \in \mathbb{F} \right\}.$$

Because

$$U = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

and the two displayed vectors are linearly independent, they form a basis of U . Hence $\dim U = 2$.

Example 28. *As a vector space over $\mathbb{R}$, the space $\mathbb{R}^2$ has dimension 2. As a vector space over $\mathbb{C}$, the space $\mathbb{C}$ has dimension 1. Although $\mathbb{R}^2$ and $\mathbb{C}$ are often identified as sets, the choice of scalar field is essential when determining dimension.*

11.3. Consequences of dimension.

Proposition 23. *If V is finite-dimensional and U is a subspace of V , then*

$$\dim U \leq \dim V.$$

Proof. A basis of U is linearly independent in V , while a basis of V spans V . The result follows from Proposition 18. $\square$

Proposition 24. *Suppose that V is finite-dimensional. Every linearly independent collection of $\dim V$ vectors in V is a basis of V .*

Proof. Write $\dim V = n$, and suppose that $v_1, \dots, v_n$ is linearly independent in V . By the basis extension theorem, this collection can be extended to a basis of V . Every basis of V has exactly n vectors, however, so no vectors need to be adjoined. Thus $v_1, \dots, v_n$ is already a basis of V . $\square$

Corollary 2. *Suppose that V is finite-dimensional and that U is a subspace of V . If $\dim U = \dim V$, then $U = V$.*

Proof. Let $u_1, \dots, u_n$ be a basis of U . Then $n = \dim U = \dim V$, and $u_1, \dots, u_n$ is linearly independent in V . By Proposition 24, it is a basis of V . Hence every vector in V is a linear combination of vectors in U , so $V \subseteq U$. The reverse inclusion holds because U is a subspace of V . Therefore $U = V$. $\square$

Example 29. *Let*

$$U = \{p \in \mathcal{P}_3(\mathbb{R}) : p'(5) = 0\}.$$

We find a basis of U . Write

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3.$$

Because

$$p'(x) = a_1 + 2a_2x + 3a_3x^2,$$

the condition $p'(5) = 0$ is equivalent to

$$a_1 + 10a_2 + 75a_3 = 0, \quad \text{or equivalently} \quad a_1 = -10a_2 - 75a_3.$$

Therefore every $p \in U$ has the form

$$\begin{aligned} p(x) &= a_0 + (-10a_2 - 75a_3)x + a_2x^2 + a_3x^3 \\ &= a_0 + a_2(x^2 - 10x) + a_3(x^3 - 75x). \end{aligned}$$

Thus the collection

$$\{1, x^2 - 10x, x^3 - 75x\}$$

spans U . It is linearly independent because its members have distinct degrees. Consequently,

$$\boxed{\{1, x^2 - 10x, x^3 - 75x\}}$$

is a basis of U , and $\dim U = 3$.

12. SEPTEMBER 21, 2026

12.1. Spanning collections and dimension.

Proposition 25. *Suppose that V is finite-dimensional. Every collection of $\dim V$ vectors that spans V is a basis of V .*

Proof. Write $\dim V = n$, and suppose that $v_1, \dots, v_n$ spans V . This collection can be reduced to a basis by removing redundant vectors. However, every basis of V has exactly n vectors, so no vectors can be removed. Thus $v_1, \dots, v_n$ is already a basis of V . $\square$

12.2. Dimension of a sum of subspaces.

Proposition 26. *If V_1 and V_2 are subspaces of a finite-dimensional vector space, then*

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

Proof. Let $v_1, \dots, v_m$ be a basis of $V_1 \cap V_2$, so that $\dim(V_1 \cap V_2) = m$. This collection is linearly independent in V_1 , so the basis extension

theorem gives a basis

$$v_1, \dots, v_m, u_1, \dots, u_j$$

of V_1 . Similarly, extend $v_1, \dots, v_m$ to a basis

$$v_1, \dots, v_m, w_1, \dots, w_k$$

of V_2 . Hence

$$\dim V_1 = m + j \quad \text{and} \quad \dim V_2 = m + k.$$

We will show that

$$v_1, \dots, v_m, u_1, \dots, u_j, w_1, \dots, w_k$$

is a basis of $V_1 + V_2$.

All the displayed vectors belong to $V_1 + V_2$, so their span is contained in $V_1 + V_2$. Conversely, every vector in $V_1 + V_2$ is a sum of a vector in V_1 and a vector in V_2 . Expressing each summand in the corresponding basis shows that the combined collection spans $V_1 + V_2$.

To prove linear independence, suppose that

$$(1) \quad a_1 v_1 + \dots + a_m v_m + b_1 u_1 + \dots + b_j u_j + c_1 w_1 + \dots + c_k w_k = 0.$$

Rearranging gives

$$c_1 w_1 + \dots + c_k w_k = -a_1 v_1 - \dots - a_m v_m - b_1 u_1 - \dots - b_j u_j.$$

The right-hand side belongs to V_1 , while the left-hand side belongs to V_2 . Therefore

$$c_1 w_1 + \dots + c_k w_k \in V_1 \cap V_2.$$

Since $v_1, \dots, v_m$ is a basis of the intersection, there are scalars $d_1, \dots, d_m$ such that

$$c_1 w_1 + \dots + c_k w_k = d_1 v_1 + \dots + d_m v_m.$$

Thus

$$c_1 w_1 + \dots + c_k w_k - d_1 v_1 - \dots - d_m v_m = 0.$$

The collection $v_1, \dots, v_m, w_1, \dots, w_k$ is a basis of V_2 and hence linearly independent. Consequently, all the c 's and d 's are zero. Equation (1)

now becomes

$$a_1v_1 + \cdots + a_mv_m + b_1u_1 + \cdots + b_ju_j = 0.$$

Because $v_1, \dots, v_m, u_1, \dots, u_j$ is a basis of V_1 , all the a 's and b 's are also zero. This proves linear independence.

The combined collection is therefore a basis of $V_1 + V_2$, and

$$\begin{aligned} \dim(V_1 + V_2) &= m + j + k \\ &= (m + j) + (m + k) - m \\ &= \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2). \end{aligned}$$

□

12.3. Linear maps on finite-dimensional spaces.

Theorem 2 (Fundamental theorem of linear maps). *Suppose that V is finite-dimensional and that $T \in \mathcal{L}(V, W)$. Then $\operatorname{im} T$ is finite-dimensional and*

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

Recall that

$$\ker T = \{v \in V : T(v) = 0\} \quad \text{and} \quad \operatorname{im} T = \{T(v) : v \in V\}.$$

Before proving the theorem, we connect the kernel and image with injectivity and surjectivity, beginning with injectivity.

Proposition 27. *Let $T \in \mathcal{L}(V, W)$. Then T is injective if and only if $\ker T = \{0\}$.*

Proof. First note that every linear map sends zero to zero, since

$$T(0) = T(0 + 0) = T(0) + T(0)$$

implies $T(0) = 0$. This does not require injectivity. In particular, $\{0\} \subseteq \ker T$.

Suppose that T is injective. If $v \in \ker T$, then

$$T(v) = 0 = T(0).$$

Injectivity gives $v = 0$, so $\ker T \subseteq \{0\}$. Therefore $\ker T = \{0\}$.

Conversely, suppose that $\ker T = \{0\}$. Let $v_1, v_2 \in V$ satisfy $T(v_1) = T(v_2)$. By linearity,

$$T(v_1 - v_2) = T(v_1) - T(v_2) = 0.$$

Hence $v_1 - v_2 \in \ker T = \{0\}$, so $v_1 = v_2$. Thus T is injective. $\square$

13. SEPTEMBER 23, 2026

13.1. Injectivity, surjectivity, and dimension. Recall from Proposition 27 that a linear map $T \in \mathcal{L}(V, W)$ is injective if and only if $\ker T = \{0\}$. In particular, if $\ker T = \{0\}$, then

$$T(v) = T(v') \implies T(v - v') = 0 \implies v - v' = 0 \implies v = v'.$$

The corresponding criterion for surjectivity follows directly from the definition of the image:

$$T \text{ is surjective} \iff \operatorname{im} T = W.$$

13.2. Proof of the fundamental theorem of linear maps. We now prove Theorem 2: if V is finite-dimensional and $T \in \mathcal{L}(V, W)$, then $\operatorname{im} T$ is finite-dimensional and

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

This identity is also called the *rank-nullity formula*; $\dim \operatorname{im} T$ is the rank and $\dim \ker T$ is the nullity.

Proof of Theorem 2. Since $\ker T$ is a subspace of the finite-dimensional space V , it has a basis $u_1, \dots, u_m$. Extend this collection to a basis

$$u_1, \dots, u_m, v_1, \dots, v_n$$

of V . Thus $\dim \ker T = m$ and $\dim V = m + n$. We claim that

$$T(v_1), \dots, T(v_n)$$

is a basis of $\operatorname{im} T$.

First, every $v \in V$ can be written as

$$v = a_1 u_1 + \dots + a_m u_m + b_1 v_1 + \dots + b_n v_n.$$

Because each u_i belongs to $\ker T$, linearity gives

$$T(v) = b_1T(v_1) + \cdots + b_nT(v_n).$$

Hence $\operatorname{im} T$ is contained in the span of the displayed images. The reverse inclusion holds because each $T(v_j)$ belongs to the subspace $\operatorname{im} T$. Therefore

$$\operatorname{im} T = \operatorname{span}\{T(v_1), \dots, T(v_n)\}.$$

To show linear independence, suppose that

$$c_1T(v_1) + \cdots + c_nT(v_n) = 0.$$

Then $T(c_1v_1 + \cdots + c_nv_n) = 0$, so $c_1v_1 + \cdots + c_nv_n \in \ker T$. There are scalars $d_1, \dots, d_m$ such that

$$c_1v_1 + \cdots + c_nv_n = d_1u_1 + \cdots + d_mu_m.$$

Rearranging gives

$$-d_1u_1 - \cdots - d_mu_m + c_1v_1 + \cdots + c_nv_n = 0.$$

The chosen basis of V is linearly independent, so all the coefficients are zero. In particular, $c_1 = \cdots = c_n = 0$.

Thus the image has a basis of n vectors, so $\dim \operatorname{im} T = n$. It follows that

$$\dim V = m + n = \dim \ker T + \dim \operatorname{im} T.$$

The same argument includes $m = 0$ or $n = 0$, using the empty basis for the zero subspace. $\square$

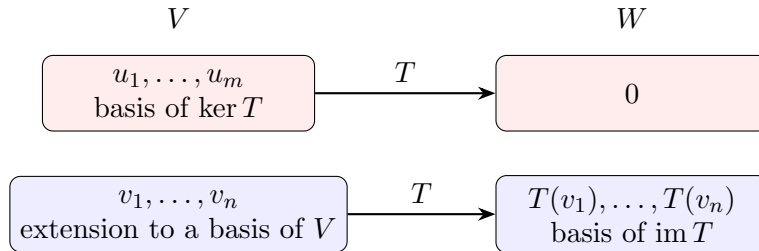


FIGURE 1. The kernel basis maps to zero; the images of the remaining basis vectors form a basis of the image.

13.3. Dimension restrictions on linear maps.

Corollary 3. *Let V and W be finite-dimensional vector spaces. If $\dim V > \dim W$, then no linear map from V to W is injective.*

Proof. If $T \in \mathcal{L}(V, W)$, then $\operatorname{im} T \subseteq W$, so $\dim \operatorname{im} T \leq \dim W$. Rank-nullity gives

$$\dim \ker T = \dim V - \dim \operatorname{im} T \geq \dim V - \dim W > 0.$$

Hence $\ker T \neq \{0\}$, and T is not injective. $\square$

Example 30. Define $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ by

$$T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \sqrt{7}x_1 + \pi x_2 + x_4 \\ 97x_1 + 3x_2 + 2x_3 \\ x_2 + 6x_3 + 7x_4 \end{bmatrix}.$$

Since $4 > 3$, this map is not injective, regardless of the particular coefficients. There is no need to solve for its kernel explicitly. The same example works over $\mathbb{C}$.

Proposition 28. *Suppose that V and W are finite-dimensional and $\dim V < \dim W$. Then no linear map from V to W is surjective.*

Proof. For $T \in \mathcal{L}(V, W)$, rank-nullity gives

$$\dim \operatorname{im} T = \dim V - \dim \ker T \leq \dim V < \dim W.$$

Thus $\operatorname{im} T \neq W$, so T is not surjective. $\square$

13.4. A complement to the kernel.

Proposition 29. *Let V be finite-dimensional, and let $T \in \mathcal{L}(V, W)$. There is a subspace $U \subseteq V$ such that*

$$V = \ker T \oplus U \quad \text{and} \quad \operatorname{im} T = \{T(u) : u \in U\}.$$

In particular, $U \cap \ker T = \{0\}$, and the restriction $T|_U: U \rightarrow \operatorname{im} T$ is bijective.

Proof. As in the proof of rank–nullity, extend a basis $u_1, \dots, u_m$ of $\ker T$ to a basis $u_1, \dots, u_m, v_1, \dots, v_n$ of V . Set

$$U = \text{span}\{v_1, \dots, v_n\}.$$

The combined basis spans V , so $V = \ker T + U$. If $w \in U \cap \ker T$, then

$$w = b_1 v_1 + \dots + b_n v_n = a_1 u_1 + \dots + a_m u_m.$$

Linear independence of the combined basis forces every coefficient to vanish. Thus $w = 0$, and the sum is direct.

Every $v \in V$ has the form $v = k + u$, where $k \in \ker T$ and $u \in U$. Therefore $T(v) = T(u)$, proving that $T|_U$ maps onto $\text{im } T$. Its kernel is $U \cap \ker T = \{0\}$, so it is also injective. $\square$

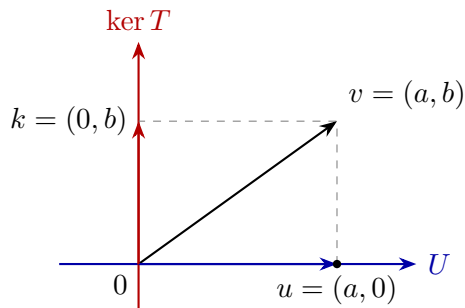


FIGURE 2. For $T: \mathbb{R}^2 \rightarrow \mathbb{R}$, $T(x, y) = x$, take U to be the horizontal axis. The decomposition $v = k + u$ gives $T(v) = T(u) = a$, while $T(k) = 0$.

14. SEPTEMBER 25, 2026

14.1. Invertible linear maps. Write Id_V for the identity map on V , defined by $\text{Id}_V(v) = v$. As with matrix multiplication, we sometimes write ST for the composition $S \circ T$.

Definition 25 (Invertibility). *A linear map $T \in \mathcal{L}(V, W)$ is invertible if there exists a linear map $S \in \mathcal{L}(W, V)$ such that*

$$S \circ T = \text{Id}_V \quad \text{and} \quad T \circ S = \text{Id}_W.$$

Such a map S is called an inverse of T .

Proposition 30 (Uniqueness of the inverse). *An invertible linear map has a unique inverse.*

Proof. Suppose $S_1, S_2 \in \mathcal{L}(W, V)$ are both inverses of T . By associativity of composition,

$$\begin{aligned} S_1 &= S_1 \circ \text{Id}_W \\ &= S_1 \circ (T \circ S_2) \\ &= (S_1 \circ T) \circ S_2 \\ &= \text{Id}_V \circ S_2 = S_2. \end{aligned}$$

□

We denote the unique inverse by $T^{-1} \in \mathcal{L}(W, V)$. Thus

$$T^{-1}T = \text{Id}_V \quad \text{and} \quad TT^{-1} = \text{Id}_W.$$

Example 31. Define $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by

$$T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -y \\ x \\ 4z \end{bmatrix}.$$

Its values on the standard basis are

$$T(e_1) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad T(e_3) = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}.$$

Hence its matrix with respect to the standard basis is

$$A = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

To recover the input from an output (x, y, z) , reverse the first two coordinates with the appropriate sign and divide the third coordinate by

4. *This gives*

$$T^{-1} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ -x \\ z/4 \end{bmatrix}, \quad A^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix}.$$

Indeed, direct multiplication gives $AA^{-1} = A^{-1}A = I_3$. Equivalently, augment A by the identity matrix and perform the row operations $R_1 \leftrightarrow R_2$, $R_2 \leftarrow -R_2$, and $R_3 \leftarrow \frac{1}{4}R_3$:

$$\left[\begin{array}{ccc|ccc} 0 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 4 & 0 & 0 & 1 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{4} \end{array} \right].$$

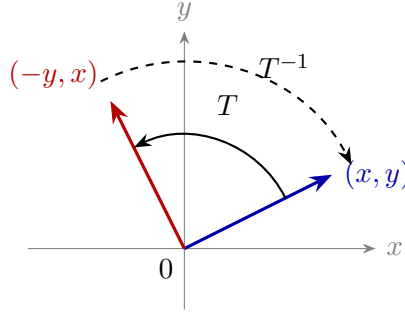


FIGURE 3. In the xy -plane, T rotates by 90° counter-clockwise and T^{-1} rotates clockwise. In the third coordinate, T multiplies by 4 and T^{-1} divides by 4.

14.2. Invertibility and bijectivity.

Proposition 31. *A linear map is invertible if and only if it is bijective.*

Proof. Let $T \in \mathcal{L}(V, W)$. Suppose first that T is invertible. If $T(u) = T(v)$, applying T^{-1} gives

$$u = T^{-1}(T(u)) = T^{-1}(T(v)) = v.$$

Thus T is injective. For every $w \in W$, the vector $v = T^{-1}(w)$ satisfies

$$T(v) = T(T^{-1}(w)) = w,$$

so T is also surjective.

Conversely, suppose that T is bijective. For each $w \in W$, surjectivity provides $v \in V$ with $T(v) = w$, and injectivity makes this vector unique. Define $S: W \rightarrow V$ by assigning to w this unique vector. By construction, $T \circ S = \text{Id}_W$. Also, for $v \in V$,

$$T(S(T(v))) = T(v),$$

so injectivity gives $S(T(v)) = v$. Thus $S \circ T = \text{Id}_V$.

It remains to check that S is linear. For $w_1, w_2 \in W$,

$$\begin{aligned} T(S(w_1) + S(w_2)) &= T(S(w_1)) + T(S(w_2)) \\ &= w_1 + w_2 \\ &= T(S(w_1 + w_2)). \end{aligned}$$

Injectivity of T yields

$$S(w_1 + w_2) = S(w_1) + S(w_2).$$

Similarly, for $\lambda \in \mathbb{F}$ and $w \in W$,

$$T(\lambda S(w)) = \lambda T(S(w)) = \lambda w = T(S(\lambda w)),$$

so $S(\lambda w) = \lambda S(w)$. Therefore $S \in \mathcal{L}(W, V)$ and S is an inverse of T . $\square$

14.3. Why the dimension hypotheses matter. In general, injectivity alone or surjectivity alone does not imply invertibility.

Example 32. On the space $\mathcal{P}(\mathbb{R})$ of all real polynomials, define

$$M: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R}), \quad (Mp)(x) = x^2 p(x).$$

This map is linear and injective: if $x^2 p(x)$ is the zero polynomial, then p is the zero polynomial. It is not surjective, because the constant polynomial 1 does not belong to its image. Thus M is not invertible, even though its domain and codomain are the same vector space.

Example 33 (A complementary example). *Differentiation*

$$D: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R}), \quad D(p) = p',$$

is surjective, since every real polynomial has a polynomial antiderivative. It is not injective, since every constant polynomial belongs to $\ker D$. Therefore D is not invertible.

Proposition 32. *Suppose that V and W are finite-dimensional with $\dim V = \dim W$, and let $T \in \mathcal{L}(V, W)$. Then the following are equivalent:*

- (i) T is invertible;
- (ii) T is injective;
- (iii) T is surjective.

Proof. By rank–nullity and the equality of dimensions,

$$\begin{aligned}
 T \text{ is injective} &\iff \ker T = \{0\} \\
 &\iff \dim \ker T = 0 \\
 &\iff \dim \operatorname{im} T = \dim V = \dim W \\
 &\iff \operatorname{im} T = W \\
 &\iff T \text{ is surjective.}
 \end{aligned}$$

Here we used the fact that a subspace of a finite-dimensional space with the same dimension is the whole space. Thus either injectivity or surjectivity implies bijectivity, which is equivalent to invertibility by Proposition 31. $\square$

14.4. One-sided inverses in equal dimensions.

Proposition 33. *Suppose that V and W are finite-dimensional with $\dim V = \dim W$. If $T \in \mathcal{L}(V, W)$ and $S \in \mathcal{L}(W, V)$, then*

$$ST = \operatorname{Id}_V \iff TS = \operatorname{Id}_W.$$

In either case, $S = T^{-1}$.

Proof. Suppose $ST = \operatorname{Id}_V$. If $v \in \ker T$, then

$$v = (ST)(v) = S(T(v)) = S(0) = 0.$$

Hence T is injective, and Proposition 32 makes it invertible. Consequently,

$$S = S(TT^{-1}) = (ST)T^{-1} = T^{-1},$$

which implies $TS = \text{Id}_W$.

Conversely, suppose $TS = \text{Id}_W$. For every $w \in W$,

$$w = T(S(w)),$$

so T is surjective and hence invertible by the same proposition. Therefore

$$S = (T^{-1}T)S = T^{-1}(TS) = T^{-1},$$

and $ST = \text{Id}_V$. □

15. SEPTEMBER 28, 2026

15.1. Isomorphic vector spaces.

Definition 26 (Isomorphism). *An isomorphism is an invertible linear map. Two vector spaces V and W over the same field $\mathbb{F}$ are isomorphic if there exists an isomorphism from V onto W . We write $V \cong W$.*

By Proposition 31, a linear map is an isomorphism precisely when it is both injective and surjective.

Proposition 34. *Two finite-dimensional vector spaces over $\mathbb{F}$ are isomorphic if and only if they have the same dimension.*

Proof. Suppose first that $T: V \rightarrow W$ is an isomorphism. Then T is injective and surjective, so $\ker T = \{0\}$ and $\text{im } T = W$. By the fundamental theorem of linear maps (Theorem 2),

$$\dim V = \dim \ker T + \dim \text{im } T = 0 + \dim W = \dim W.$$

Conversely, let $\dim V = \dim W = n$. Choose bases $v_1, \dots, v_n$ of V and $w_1, \dots, w_n$ of W . By the linear map lemma (Lemma 2), there is a unique linear map $T: V \rightarrow W$ satisfying $T(v_j) = w_j$ for all j . Explicitly,

$$T(c_1v_1 + \dots + c_nv_n) = c_1w_1 + \dots + c_nw_n.$$

This map is well-defined because coordinates with respect to a basis are unique. Since $w_1, \dots, w_n$ spans W , every vector in W occurs as an output, so T is surjective. If $T(c_1v_1 + \dots + c_nv_n) = 0$, then

$$c_1w_1 + \dots + c_nw_n = 0.$$

Linear independence of the w_j gives $c_1 = \dots = c_n = 0$. Thus $\ker T = \{0\}$, and Proposition 27 shows that T is injective. Therefore T is an isomorphism. When $n = 0$, the unique map between the two zero spaces is an isomorphism as well. $\square$

15.2. The space of linear maps and the space of matrices. Fix ordered bases

$$\mathcal{B} = (v_1, \dots, v_n) \quad \text{and} \quad \mathcal{C} = (w_1, \dots, w_m)$$

of V and W , respectively. To keep track of both bases, write $[T]_{\mathcal{C}}^{\mathcal{B}}$ for the matrix of $T \in \mathcal{L}(V, W)$, characterized by

$$[T(v)]_{\mathcal{C}} = [T]_{\mathcal{C}}^{\mathcal{B}}[v]_{\mathcal{B}}.$$

Thus the j th column is $[T(v_j)]_{\mathcal{C}}$, as in Theorem 1. The superscript specifies the domain basis and the subscript specifies the codomain basis. Thus the matrix takes $\mathcal{B}$ -coordinates to $\mathcal{C}$ -coordinates.

Proposition 35. *The map*

$$\mathcal{M}: \mathcal{L}(V, W) \longrightarrow M_{m \times n}(\mathbb{F}), \quad \mathcal{M}(T) = [T]_{\mathcal{C}}^{\mathcal{B}},$$

is an isomorphism of vector spaces.

Proof. The calculations in Section 7.1 show that

$$\mathcal{M}(T + S) = \mathcal{M}(T) + \mathcal{M}(S), \quad \mathcal{M}(\alpha T) = \alpha \mathcal{M}(T).$$

Hence $\mathcal{M}$ is linear.

For injectivity, suppose that $\mathcal{M}(T)$ is the zero matrix. Every column is zero, so for each j ,

$$T(v_j) = 0w_1 + \dots + 0w_m = 0.$$

Since the v_j form a basis, linearity implies that $T(v) = 0$ for every $v \in V$. Thus T is the zero map, and $\ker \mathcal{M} = \{0\}$. By Proposition 27, $\mathcal{M}$ is injective.

For surjectivity, take an arbitrary matrix $A = [A_{ij}] \in M_{m \times n}(\mathbb{F})$. Prescribe

$$T(v_j) = \sum_{i=1}^m A_{ij} w_i \quad (j = 1, \dots, n).$$

Lemma 2 gives a linear map $T \in \mathcal{L}(V, W)$ with these values, and its matrix is exactly A . Therefore $\mathcal{M}$ is surjective. Being linear and bijective, it is an isomorphism by Proposition 31. $\square$

Corollary 4. *If V and W are finite-dimensional, then $\mathcal{L}(V, W)$ is finite-dimensional and*

$$\dim \mathcal{L}(V, W) = (\dim V)(\dim W).$$

Proof. Set $n = \dim V$ and $m = \dim W$. For $1 \leq i \leq m$ and $1 \leq j \leq n$, let E_{ij} be the matrix with a 1 in entry (i, j) and zeros elsewhere. Every matrix $A = [A_{ij}]$ has the unique expansion

$$A = \sum_{i=1}^m \sum_{j=1}^n A_{ij} E_{ij}.$$

Thus the mn matrices E_{ij} form a basis of $M_{m \times n}(\mathbb{F})$. Because $\mathcal{M}$ is a linear isomorphism, the maps $\mathcal{M}^{-1}(E_{ij})$ form a basis of $\mathcal{L}(V, W)$. Hence this space is finite-dimensional with dimension mn . If either m or n is zero, the space of linear maps is the zero space, and the same formula holds. $\square$

15.3. Invertible matrices.

Definition 27 (Invertible matrix). *A square matrix $A \in M_{n \times n}(\mathbb{F})$ is invertible if there exists $B \in M_{n \times n}(\mathbb{F})$ such that*

$$AB = BA = I_n,$$

where I_n is the identity matrix. We call B the inverse of A and denote it by A^{-1} .

The inverse is unique: if B and C are both inverses of A , then

$$B = B(AC) = (BA)C = C.$$

Let V and W be finite-dimensional, and let $T: V \rightarrow W$ be an isomorphism. By Proposition 34, both spaces have the same dimension n . Let

$$A = [T]_{\mathcal{C}}^{\mathcal{B}}, \quad B = [T^{-1}]_{\mathcal{B}}^{\mathcal{C}}.$$

By Section 7.2, composition corresponds to matrix multiplication. Thus $T^{-1}T = \text{Id}_V$ and $TT^{-1} = \text{Id}_W$ give

$$BA = I_n \quad \text{and} \quad AB = I_n.$$

Consequently,

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = ([T]_{\mathcal{C}}^{\mathcal{B}})^{-1}.$$

Notice that the roles of the domain and codomain bases are reversed for the inverse map.

15.4. Change of basis. Let

$$\mathcal{B} = (v_1, \dots, v_n) \quad \text{and} \quad \mathcal{C} = (u_1, \dots, u_n)$$

be two ordered bases of the same vector space V . Define

$$P = [\text{Id}_V]_{\mathcal{C}}^{\mathcal{B}}, \quad Q = [\text{Id}_V]_{\mathcal{B}}^{\mathcal{C}}.$$

Although the identity map leaves each vector unchanged, its matrix need not be I_n when the domain and codomain use different bases. Here

$$[v]_{\mathcal{C}} = P[v]_{\mathcal{B}}, \quad [v]_{\mathcal{B}} = Q[v]_{\mathcal{C}}.$$

In particular, the columns of P are the old basis vectors expressed in the new basis:

$$P = \begin{bmatrix} | & & | \\ [v_1]_{\mathcal{C}} & \cdots & [v_n]_{\mathcal{C}} \\ | & & | \end{bmatrix}.$$

Proposition 36. *The matrices P and Q are invertible, and each is the inverse of the other.*

Proof. Changing from $\mathcal{B}$ -coordinates to $\mathcal{C}$ -coordinates and then back represents the composition $\text{Id}_V \circ \text{Id}_V = \text{Id}_V$ with $\mathcal{B}$ used at both ends. Hence $QP = I_n$. Reversing the order gives $PQ = I_n$. Thus $Q = P^{-1}$. $\square$

Proposition 37 (Change-of-basis formula). *Let $T \in \mathcal{L}(V, V)$, and let*

$$A = [T]_{\mathcal{B}}^{\mathcal{B}}, \quad B = [T]_{\mathcal{C}}^{\mathcal{C}}.$$

With $P = [\text{Id}_V]_{\mathcal{C}}^{\mathcal{B}}$, we have

$$B = PAP^{-1} = PAQ.$$

Proof. Applying T and then changing coordinates has the same effect as changing coordinates first and then applying T in the new basis. The following diagram expresses this equality:

$$\begin{array}{ccc} [v]_{\mathcal{B}} & \xrightarrow{A} & [T(v)]_{\mathcal{B}} \\ P \downarrow & & \downarrow P \\ [v]_{\mathcal{C}} & \xrightarrow{B} & [T(v)]_{\mathcal{C}}. \end{array}$$

Equivalently, the identity $T \circ \text{Id}_V = \text{Id}_V \circ T$ gives

$$BP = PA.$$

Multiplying on the right by $P^{-1} = Q$ yields $B = PAP^{-1} = PAQ$. $\square$

To interpret the formula, start with a vector in $\mathcal{C}$ -coordinates. The rightmost factor P^{-1} converts it to $\mathcal{B}$ -coordinates, A computes the coordinates of its image under T , and P converts the result back to $\mathcal{C}$ -coordinates.